matrix-grid · rendering jank

The grid that moved after it painted.

`matrix-grid` drew its columns from its own cell text. When the web font landed a frame after first paint, every column re-solved and every cell slid sideways — with nothing overflowing for a fit gate to catch.

What was moving

Column widths were a function of the font that had not loaded yet.

Measured, not inferred

Same deck, same Studio, fonts blocked against fonts allowed. Blocked, the columns never moved.

The five columns moved

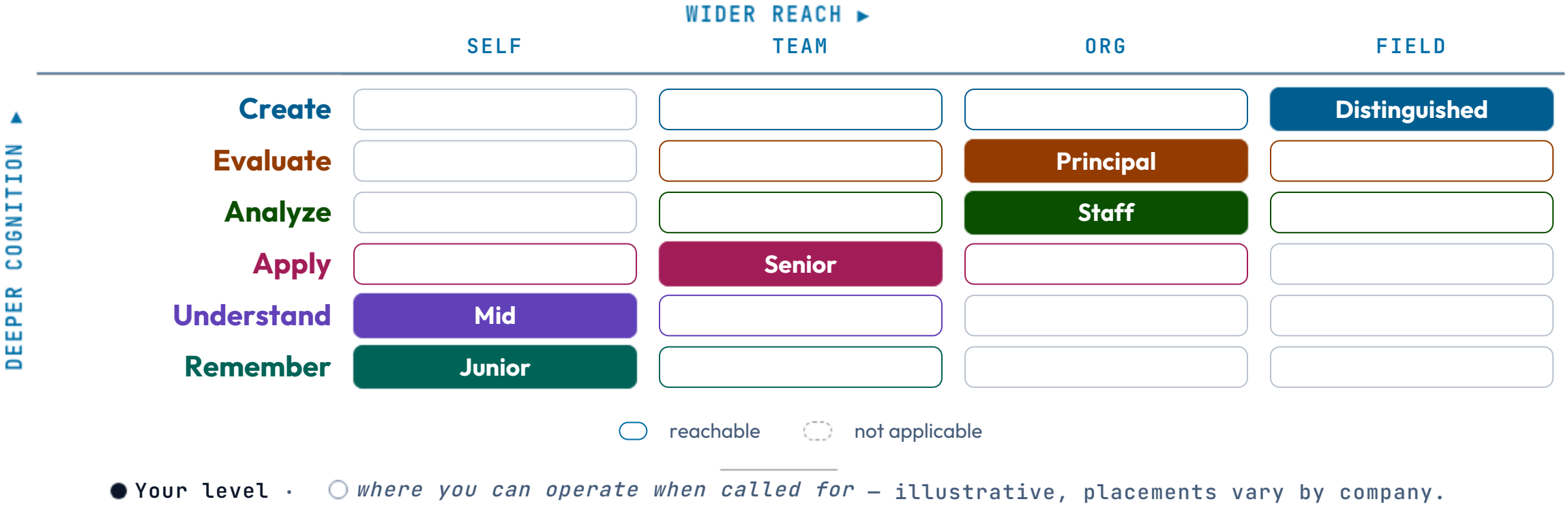
`258.2 · 165.7 · 175.5 · 216.4 · 304.2` on first paint. `248.7 · 176.9 · 178.9 · 218.9 · 296.6` one frame later.

Invisible to every gate

The table was 1120px wide before and after. Nothing overflowed, so nothing was reported.

Your level is a cell, not a rung.

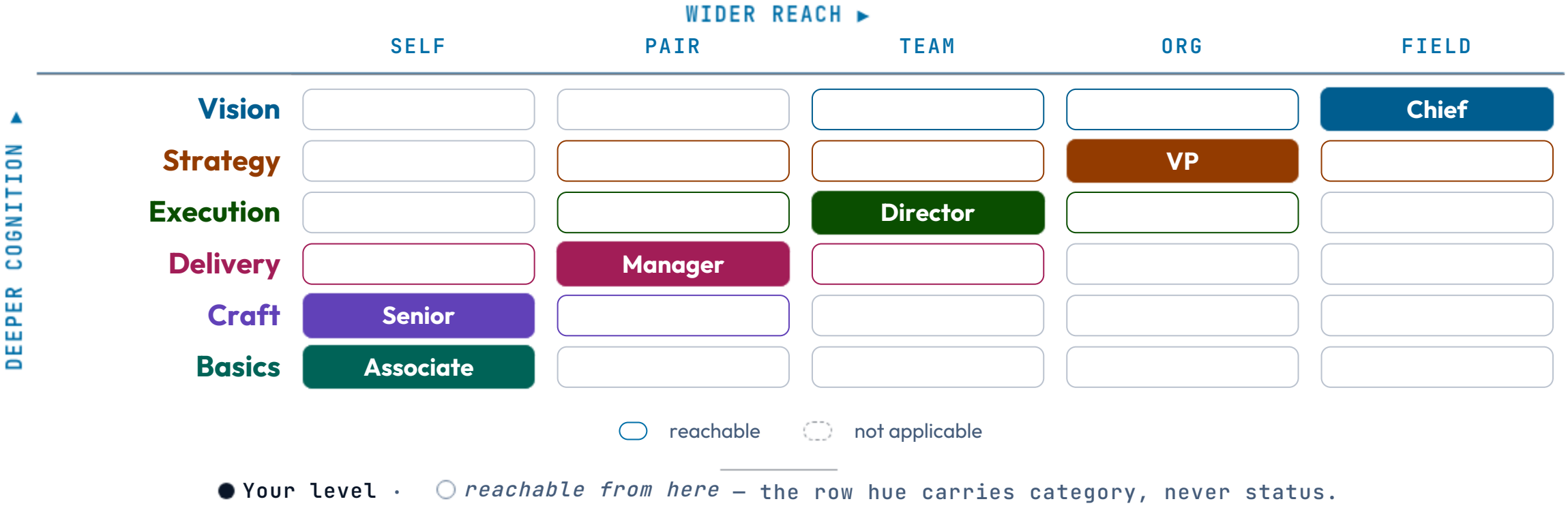
Every column is the same width, so the reach axis reads as equal steps.



● Your level · ○ *where you can operate when called for* — illustrative, placements vary by company.

Six categories by five columns is the grid's practical ceiling.

The widest label no longer buys width from its neighbors.



How it is held still

`table-layout: fixed` takes the text out of the geometry.

Geometry stops reading text

A fixed table divides its width by declaration. A late font changes the glyphs, never the column.

The sibling already did it

`roadmap`, the other table in this bucket, has been fixed-layout all along, for this reason.

Equal steps read truer

Under `auto` the reach axis was sized by the longest label: 165.7px against 304.2px, an 1.8x spread.

Only wide is wide enough

Every narrower family still has to negotiate.

Measured as text advance

`scrollWidth` on a centered flex box misses the half that overflows left. A Range over the text node is the honest unit — and it reads `-25.2px` where `scrollWidth` reported a 2px rounding artifact.

Square fails like portrait, later

`--canvas-scale` raises the type as the box narrows. `Distinguished` clears its cell by `+75.9px` at wide, misses by `-25.2px` at square and `-44.5px` under `mode: sketch`.

Only wide is wide enough

And the type has nowhere left to go.

The type floor is reached

`--fs-meta` is the floor of the twelve-token scale, so the step the narrow arm already takes is the last one available.

So only wide takes fixed

Square, tall and strip keep `auto` and render byte-identically. The shift stays live there, and `scale-2xl` leaves wide only `+19.7px`.

Related

See also.

`roadmap` — the sibling table, fixed-layout since it shipped

`obligation-matrix` — rows × columns of pass/partial/exempt status

`engineering/jank.md` — does the layout stay put as content grows